

NITHISH B

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EDUCATION

R.V. College of Engineering

Bachelor of Engineering in Computer Science and Engineering (Cybersecurity) | CGPA: 8.73

Bengaluru, Karnataka

Jun 2023 – Present

Vivekananda Pre-University College

Physics, Chemistry, Mathematics, Biology – 95.66%

Puttur, Dakshina Kannada, Karnataka

2022 – 2023

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL

Core Computer Science: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS

Cybersecurity: Basics of Ethical Hacking, Applied Cryptography, Blockchain

AI/ML & Data: Machine Learning (Scikit-learn, Pandas, NumPy)

EXPERIENCE

Centre of Excellence in Computer Vision

August 2026 – Present

Research Intern — *PrivWatch: Privacy-Preserving AI-Based Surveillance System*

- Analyze real-time video streams by extracting and temporarily buffering frames in RAM; apply AI and deep-learning techniques for activity and anomaly detection, classify 2 critical event types (fights and collapses), and generate real-time alerts.
- Design and compare conventional and privacy-preserving surveillance pipelines; automatically delete 100% of processed frames after inference, retaining 0 frames in persistent storage and demonstrating the benefits of transient video processing.

PROJECTS

Shiksha Sahayak: RAG-Based Educational AI Assistant with LLM Integration | [🔗 Link](#)

Team Lead

- Created a RAG-based educational assistant using Python, Flask, Groq LLM, LangChain, and TF-IDF for context-aware academic responses with PDF-based knowledge retrieval.
- Built an end-to-end PDF ingestion and retrieval pipeline using PyPDF2, LangChain text splitting with 1,000-character chunks and 200-character overlap, scikit-learn TF-IDF vector indexing, and minimum similarity score filtering; deployed it with Docker and Gunicorn.

CarbonChain — AI-Powered Carbon Credit Verification Platform | [🔗 Link](#)

Developer

- Engineered a decentralized carbon credit verification platform using React, TypeScript, REST APIs, Hardhat, and Solidity smart contracts for secure telemetry and blockchain-based verification.
- Built a real-time compliance dashboard and block explorer to track energy consumption, CO₂ emissions, carbon credits, cryptographic hashes, and blockchain transactions.

Sentinel — Adaptive & Continuous Intrusion Detection System | [🔗 Link](#)

Developer

- Implemented a full-stack continuous authentication system using Isolation Forest anomaly detection on behavioral telemetry, with 37-feature parity, RobustScaler preprocessing, and calibrated trust scoring.
- Implemented a Flask REST API with EMA-based trust smoothing, fail-closed heartbeat decay, and Argon2id-secured credentials and recovery keys stored in SQLite.

FootPrint — Hybrid DFT-ML Pipeline for Piezoelectric Voltage Prediction and IoT Energy Harvesting | [🔗 Link](#)

Team Lead

- Built a hybrid DFT-guided ML pipeline using Gaussian simulation outputs, Mulliken charge analysis, multipole moment calculations, and MATLAB post-processing to train a gradient boosting model for piezoelectric voltage prediction.
- Designed a FastAPI backend with real-time inference and a JavaScript frontend with Chart.js visualizations for live prediction tracking and voltage scaling, supporting IoT-based monitoring of ZnO and Fe-doped ZnO energy harvesting systems.

PUBLICATIONS

A Secure and Decentralized Framework for Intellectual Property Protection Using Blockchain and Cryptographic Techniques

- *International Research Journal of Modernization in Engineering Technology and Science (IRJMETs)*, Vol. 08, Issue 06, June 2026.
- Proposed a decentralized framework based on blockchain and cryptography to enhance the security, integrity, and protection of intellectual property.

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification — *DeepLearning.AI & Stanford Online (Coursera)*

Data Science for Engineers — NPTEL

Knowledge Management — NPTEL

Implement Real-Time Face Detection with OpenCV & Python — EDUCBA